

Elementary Problems Diophantine Equation Solutions

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Basic Work

1. We may simply take $m = 3k^2$, $n = 1$ and $p = 3k$ for some positive integer k so that

$$4mn - m - n = 4(3k^2)(1) - 3k^2 - 1 = 9k^2 - 1 = p^2 - 1.$$

2. The equation is the same as

$$(m - 1)(n + 2) = 1.$$

Thus, $m - 1 = n + 2 = \pm 1$. The solutions are $(m, n) = (2, -1), (0, -3)$. Note that the steps are reversible.

3. The equation is the same as $(2m - 3)\left(n + \frac{3}{2}\right) = \frac{1}{2}$, i.e.

$$(2m - 3)(2n + 3) = 1.$$

Thus, $2m - 3 = 2n + 3 = \pm 1$. The solutions are $(m, n) = (2, -1), (1, -2)$.

4. The equation is the same as $(x - p)(y - p) = p^2$. As p is a prime, this yields

$$(x - p, y - p) = (\pm 1, \pm p^2), (\pm p, \pm p), (\pm p^2, \pm 1).$$

Solving these, we have

$$(x, y) = (p \pm 1, p \pm p^2), (2p, 2p), (0, 0), (p \pm p^2, p \pm 1).$$

Since the steps are reversible, these are all solutions.

5. We apply the Euclidean algorithm. Firstly, we find that

$$\begin{aligned} 107 &= 2(43) + 21, \\ 43 &= 2(21) + 1. \end{aligned}$$

Therefore, we have

$$\begin{aligned} 1 &= 43 - 2(21) \\ &= 43 - 2(107 - 2(43)) \\ &= (1 + 2 \times 2)(43) - 2(107) \\ &= 5(43) - 2(107). \end{aligned}$$

It follow that $20(43) - 8(107) = 4$. This means $(m, n) = (-8, 20)$ is a solution. As $(107, 43) = 1$, the general solution is given by

$$m = -8 + 43k \quad \text{and} \quad n = 20 - 107k$$

where k is any integer.

6. The equation is the same as

$$(2m - n)(2m + n) = 35.$$

As $2m + n > 2m - n$, we have

$$(2m - n, 2m + n) = (1, 35), (5, 7).$$

The solutions are $(m, n) = (9, 17), (3, 1)$.

7. Let $n^2 + 12n + 8 = m^2$ for some $m \in \mathbb{N}$. By completing square, we get

$$(n + 6)^2 - 28 = m^2.$$

This is the same as $(n + 6 - m)(n + 6 + m) = 28$. As $n + 6 + m > n + 6 - m$ and they have the same parity, we have

$$(n + 6 - m, n + 6 + m) = (2, 14).$$

This yields $(m, n) = (6, 2)$. Therefore, n can only be 2.

8. If both roots are integers, then the sum of roots $-p$ is an integer. This shows p is an integer. Note that we need $\Delta = p^2 - 12p = k^2$ for some integer k . This can be rewritten as $(p - 6)^2 - 36 = k^2$, i.e. $(p - 6)^2 - k^2 = 36$. Hence, we have

$$(p - 6 - k)(p - 6 + k) = 36.$$

Note that $p - 6 - k$ and $p - 6 + k$ have the same parity. The only possibilities are

$$(p - 6 - k, p - 6 + k) = (\pm 2, \pm 18), (\pm 6, \pm 6), (\pm 18, \pm 2).$$

Solving for p , we can only have $p = 16, -4, 12, 0$. We check that

$$x^2 + 16x + 48 = 0 \Rightarrow x = -4, -12,$$

$$x^2 - 4x - 12 = 0 \Rightarrow x = 6, -2,$$

$$x^2 + 12x + 36 = 0 \Rightarrow x = -6, -6,$$

$$x^2 = 0 \Rightarrow x = 0, 0.$$

9. Let $(m, n) = d$, $m = da$ and $n = db$ so that $(a, b) = 1$ and $[m, n] = dab$. The equation becomes

$$dab + d = da + db.$$

This can be rewritten as $d(a - 1)(b - 1) = 0$. Thus, we have $a = 1$ or $b = 1$. WLOG assume $a = 1$. Then $m = d$ divides $n = db$ as desired.

10. Note that

$$\begin{aligned} 30 &= m^2n + n^2 + m - mn^2 - m^2 - n \\ &= (m - n)(mn - m - n + 1) \\ &= (m - n)(m - 1)(n - 1). \end{aligned}$$

This shows $m > n$. If none of $m - n$, $m - 1$ and $n - 1$ is 1, then they must be 2, 3, 5 in some order. Noting that

$$(m - 1) - (n - 1) = m - n,$$

we must have $m - 1 = 5$ and $n - 1 = 2, 3$. This gives the solutions $(m, n) = (6, 3), (6, 4)$. Note that the steps are reversible.

If $m - n = 1$, then $30 = (m - 1)(n - 1) = n(n - 1)$. Clearly, the only solution is $n = 6$ and $m = 7$.

If $n - 1 = 1$, then $n = 2$ and $30 = (m - 2)(m - 1)$. Clearly, the only solution is $m = 7$. Therefore, the solutions are $(m, n) = (6, 3), (6, 4), (7, 6), (7, 2)$.

11. Let $p - 1 = m^2$ and $2p - 1 = n^2$ for some $m, n \in \mathbb{N}$. Then we have

$$n^2 - m^2 = (2p - 1) - (p - 1) = p.$$

This can be factorized as $(n - m)(n + m) = p$. As p is a prime, this yields $n - m = 1$ and $n + m = p$. Thus, $m = \frac{p - 1}{2}$. From $p - 1 = m^2$, we need to solve

$$p - 1 = \left(\frac{p - 1}{2}\right)^2.$$

As $p > 1$, this yields $1 = \frac{p - 1}{4}$, i.e. $p = 5$. We check that both $5 - 1 = 4$ and $2 \times 5 - 1 = 9$ are perfect squares. Hence, the only solution is $p = 5$.

12. Let $\sqrt{n} + \sqrt{n + 2005} = m$. Then we have

$$n + 2005 = (m - \sqrt{n})^2 = m^2 + n - 2m\sqrt{n}.$$

This implies $2m\sqrt{n} = m^2 - 2005$. Thus, $\sqrt{n}$ is a rational number. As n is an integer, $\sqrt{n}$ has to be an integer. Therefore,

$$2m \mid m^2 - 2005.$$

This yields $m \mid 2005$ so that $m = 1, 5, 401, 2005$. In order that $\sqrt{n} = \frac{m^2 - 2005}{2m} > 0$. The only possibilities are $m = 401, 2005$. Correspondingly, we get $n = 198^2 = 39204$ and $n = 1002^2 = 1004004$. One checks that these are the solutions.

13. For $n = 1$, we can take any positive integer. For $n \geq 2$, we take $1, 1, \dots, 1, 2, n$ where there are $n - 2$ copies of 1's. Then

$$1 + 1 + \dots + 1 + 2 + n = 2n = (1)(1) \cdots (1)(2)(n).$$

14. Consider any $n > ab - a - b$. As $(a, b) = 1$, there exist $x, y \in \mathbb{Z}$ such that $ax + by = n$. We pick the pair (x, y) for which x is a nonnegative integer and is the smallest possible. We claim that $x \leq b - 1$. Indeed, if $x \geq b$, then we can write

$$n = ax + by = a(x - b) + b(y + a).$$

This shows the pair $(x - b, y + a)$ has the first entry $x - b \geq 0$, which is smaller than x . This contradicts the minimality of x . Having $x \leq b - 1$, we can easily prove that $y \geq 0$. Indeed, if $y \leq -1$, then

$$n = ax + by \leq a(b - 1) + b(-1) = ab - a - b.$$

This is a contradiction.

Next, we show that $ab - a - b$ cannot be represented in the desired form. Suppose $ab - a - b = ax + by$ for some $x, y \geq 0$. This can be rewritten as $(b - 1 - x)a = (y + 1)b$. This implies $a \mid (y + 1)b$. As $(a, b) = 1$, this yields $a \mid y + 1$. Let $y = \ell a - 1$ for some integer ℓ . Note that $\ell \geq 1$ since y is nonnegative. Therefore, $y \geq a - 1$. By symmetry, we have $x \geq b - 1$. Then

$$ab - a - b = ax + by \geq a(b - 1) + b(a - 1) = 2ab - a - b.$$

This is a contradiction. Hence, the largest number not expressible in the form $ax + by$ is $ab - a - b$.

In general, we claim that $N(k) = kab - a - b$. Firstly, consider any $n > N(k)$. From above, we can pick a solution to $ax + by = n$ with $0 \leq x \leq b - 1$. Then we get the solutions

$$(x, y), (x + b, y - a), (x + 2b, y - 2a), \dots, (x + (k - 1)b, y - (k - 1)a).$$

To show that these are nonnegative solutions, it suffices to show $y - (k - 1)a \geq 0$. Suppose on the contrary that $y \leq (k - 1)a - 1$. Then

$$n = ax + by \leq a(b - 1) + b((k - 1)a - 1) = kab - a - b.$$

This is a contradiction.

Next, we show that $N(k)$ cannot be represented in the desired form in k ways. As above, consider $kab - a - b = ax + by$ for some $x, y \geq 0$. This can be rewritten as $(kb - 1 - x)a = (y + 1)b$. Thus, $a \mid y + 1$. Let $y = \ell a - 1$ for some integer ℓ . If there are k solutions, then there exists one of them with $\ell \geq k$. Then $y \geq ka - 1$. As above, we have $x \geq b - 1$. Then

$$kab - a - b = ax + by \geq a(b - 1) + b(ka - 1) = (k + 1)ab - a - b.$$

This is a contradiction. Hence, $N(k) = kab - a - b$.

15. This is a Pell's equation. The smallest positive integer solution is $(m, n) = (2, 1)$. The general solution (m_k, n_k) is given by

$$m_k + n_k\sqrt{3} = (2 + \sqrt{3})^k$$

where $k \in \mathbb{N}$. Note that

$$m_{k+1} + n_{k+1}\sqrt{3} = (m_k + n_k\sqrt{3})(2 + \sqrt{3}) = (2m_k + 3n_k) + (m_k + 2n_k)\sqrt{3}.$$

This gives the recurrence relations

$$\begin{aligned} m_{k+1} &= 2m_k + 3n_k, \\ n_{k+1} &= m_k + 2n_k. \end{aligned}$$

Combining these, we have

$$m_k = n_{k+1} - 2n_k = \frac{1}{3}(m_{k+2} - 2m_{k+1}) - \frac{2}{3}(m_{k+1} - 2m_k),$$

i.e.

$$m_{k+2} = 4m_{k+1} - m_k.$$

The characteristic equation is $\lambda^2 - 4\lambda + 1 = 0$. Its roots are $2 \pm \sqrt{3}$. Therefore, $m_k = A(2 + \sqrt{3})^k + B(2 - \sqrt{3})^k$ for some constants A, B . The initial values are $m_1 = 2$ and $m_2 = 7$. Solving the system

$$\begin{cases} (2 + \sqrt{3})A + (2 - \sqrt{3})B = 2, \\ (2 + \sqrt{3})^2A + (2 - \sqrt{3})^2B = 7, \end{cases}$$

we obtain $A = \frac{1}{2}$ and $B = \frac{1}{2}$. Hence,

$$m_k = \frac{1}{2}((2 + \sqrt{3})^k + (2 - \sqrt{3})^k).$$

From this we have

$$n_k = \frac{1}{3}(m_{k+1} - 2m_k) = \frac{\sqrt{3}}{6}((2 + \sqrt{3})^k - (2 - \sqrt{3})^k).$$

Hence, the solutions are

$$(m, n) = \left(\frac{1}{2}((2 + \sqrt{3})^k + (2 - \sqrt{3})^k), \frac{\sqrt{3}}{6}((2 + \sqrt{3})^k - (2 - \sqrt{3})^k) \right).$$

16. Let $a = m + n$ and $b = mn$. Then we have

$$m^2(n - 1) + n^2(m - 1) = mn(m + n) - (m^2 + n^2) = ab - a^2 + 2b.$$

We need to solve $ab - a^2 + 2b = 1$. This implies $b = \frac{a^2 + 1}{a + 2}$ since it is obvious that $a \neq -2$. From $a + 2 \mid a^2 + 1$, we have $a + 2 \mid (-2)^2 + 1 = 5$. The only possibilities are $a + 2 = \pm 1, \pm 5$, i.e. $a = -7, -3, -1, 3$. This yields

$$(a, b) = (-7, -10), (-3, -10), (-1, 2), (3, 2).$$

From $m + n = a$ and $mn = b$, we have

$$m - n = \pm \sqrt{(m + n)^2 - 4mn} = \pm \sqrt{a^2 - 4b}.$$

So we can solve for m and n for each pair of a and b . We find that the only integer solutions are

$$(m, n) = (2, -5), (-5, 2), (2, 1), (1, 2).$$

17. After expansion and simplification, we get

$$a^2b^2 + 3a^2b + ab + 2b^3 = 3ab^2.$$

As b is nonzero, this implies

$$2b^2 + (a^2 - 3a)b + (3a^2 + a) = 0.$$

Thus,

$$b = \frac{-a^2 + 3a \pm \sqrt{(a^2 - 3a)^2 - 4(2)(3a^2 + a)}}{4}. \quad (1)$$

The discriminant is

$$(a^2 - 3a)^2 - 4(2)(3a^2 + a) = a^4 - 6a^3 - 15a^2 - 8a = a(a + 1)^2(a - 8).$$

As b is an integer, we must have $a(a - 8) = k^2$ for some $k \in \mathbb{N}$. This can be rewritten as $(a - 4)^2 - k^2 = 16$, i.e.

$$(a - 4 - k)(a - 4 + k) = 16.$$

Note that $a - 4 - k$ and $a - 4 + k$ have the same parity. Thus,

$$(a - 4 - k, a - 4 + k) = (\pm 2, \pm 8), (\pm 4, \pm 4), (\pm 8, \pm 2).$$

Solving these, we find that $a = -1, 8, 9$ as a is nonzero. From (1), we find that the solutions are

$$(a, b) = (-1, -1), (8, -10), (9, -6), (9, -21).$$

No checking is required as (1) is equivalent to the given equation.

18. All positive integers except 1, 3 satisfy the condition.

Firstly, for even n , we take

$$x_1 = \cdots = x_{\frac{n}{2}} = 1 \quad \text{and} \quad x_{\frac{n}{2}+1} = \cdots = x_n = -1$$

so that $x_1 + \cdots + x_n = 0$ and $x_1^2 + \cdots + x_n^2 = n$. So we can take $y = 1$.

For $n = 1$, we have $x_1 = 0$, which contradicts the nonzero condition.

For $n = 3$, we may assume $\gcd(x_1, x_2, x_3) = 1$ since otherwise we can divide each of them by the greatest common divisor to obtain the same system of Diophantine equations. In particular, we may assume x_1 is odd. The equations combine to give

$$x_1^2 + x_2^2 + (-x_1 - x_2)^2 = 3y^2,$$

i.e.

$$2(x_1^2 + x_1x_2 + x_2^2) = 3y^2.$$

Note that y is even. Thus, both sides are divisible by 4. However, we have

$$x_1^2 + x_1x_2 + x_2^2 \equiv 1 + x_2 + x_2^2 \equiv 1 \pmod{2}$$

no matter x_2 is odd or even. Thus, $2(x_1^2 + x_1x_2 + x_2^2)$ cannot be a multiple of 4. This is a contradiction. Hence, there is no solution when $n = 3$.

For odd $n \geq 5$, we take

$$x_1 = x_2 = x_3 = x_4 = 1, \quad x_5 = -4, \quad x_6 = 2, \quad x_7 = -2, \quad x_8 = 2, \quad x_9 = -2, \dots$$

so that $x_1 + \cdots + x_n = 0$ and

$$x_1^2 + \cdots + x_n^2 = 20 + 4(n - 5) = n \times 2^2.$$

So we can take $y = 2$.

Therefore, n can be any positive integer except 1 and 3.

19. (a) True. We have

$$\begin{aligned} (12345, 13579) &= (12345, 13579 - 12345) = (12345, 1234) \\ &= (12345 - 1234 \times 10, 1234) = (5, 1234) = 1. \end{aligned}$$

Therefore, the equation has infinitely many integer solutions, which can be found via the Euclidean algorithm.

(b) True. We have

$$11111 = 4m^2 - n^2 = (2m - n)(2m + n).$$

There are finitely many factorizations of 11111 as a product of two integers. For each factorization, there is at most one solution in m and n . Indeed, if $2m - n = a$ and $2m + n = b$, then we get $m = \frac{a+b}{4}$ and $\frac{b-a}{2}$, which may or may not be integers.

- (c) False. We choose a and b such that $m = a + 2b$ and $n = a + 3b$. In other words, we let $a = 3m - 2n$ and $b = n - m$. Then the equation becomes

$$3(a + 2b)^2 - 2(a + 3b)^2 = 1.$$

This is the same as $a^2 - 6b^2 = 1$. This is a Pell's equation, which has infinitely many solutions in a and b . For each such pair (a, b) , we get a distinct pair (m, n) . Hence, there are infinitely many solutions.

- (d) True. Solving the equation $m^2 + f(n)m + g(n) = 0$, we get

$$m = \frac{-f(n) \pm \sqrt{f(n)^2 - 4g(n)}}{2}.$$

If this is an integer, then $\sqrt{f(n)^2 - 4g(n)}$ must be an integer. It follows that $f(n)^2 - 4g(n)$ is a perfect square.

Conversely, if $f(n)^2 - 4g(n) = k^2$ for some integer k , then we have

$$m = \frac{-f(n) \pm k}{2}.$$

Note that k has the same parity as $f(n)$ as $k^2 \equiv f(n)^2 \pmod{2}$. It follows that m is an integer. In other words, the equation has an integer solution.

Thus, the equivalence is established.

- (e) False. For example, the equation $2m + 2n = 1$ has no integer solution since the left-hand side is even while the right-hand side is odd. However, the linear congruence equation $2m + 2n \equiv 1 \pmod{p}$ has a solution for any prime $p \neq 2$ since $(p, 2) = 1$ (for example, we may take $m = \frac{p+1}{2}$ and $n = 0$).

Congruence

20. By considering modulo 3, we have $0 \equiv 1 \pmod{3}$. This is a contradiction. Hence, there is no solution.
21. Suppose on the contrary that such integers exist. Clearing denominators, we have

$$a_2 a_3 \cdots a_{2k} + a_1 a_3 a_4 \cdots a_{2k} + \cdots + a_1 a_2 \cdots a_{2k-1} = a_1 a_2 \cdots a_{2k}.$$

As each a_j is odd, this implies

$$1 + 1 + \cdots + 1 \equiv 1 \pmod{2}.$$

However, the left-hand side consists of $2k$ ones. This yields $0 \equiv 1 \pmod{2}$, which is impossible.

Thus, there are no such integers.

22. Consider $n = a^2 - b^2 = (a - b)(a + b)$. Note that $a - b$ and $a + b$ have the same parity. If both are odd, then n is odd. If both are even, then n is a multiple of 4. Thus, n cannot be of the form $4k + 2$.

For odd $n = 2k - 1$, we can take $a - b = 1$ and $a + b = 2k - 1$. This gives the representation $n = k^2 - (k - 1)^2$.

For $n = 4k$, we can take $a - b = 2$ and $a + b = 2k$. This gives the representation $n = (k + 1)^2 - (k - 1)^2$.

Hence, n is any integer not of the form $4k + 2$.

23. If $m \geq 7$, we consider modulo 7 to get $n^2 \equiv 6 \pmod{7}$. This is impossible since a square number can only be congruent to 0, 1, 2, 4 modulo 7.

By checking $m = 1, 2, \dots, 6$, we find that the only solutions are $(m, n) = (4, 10), (5, 14)$.

24. When $m = 1$ and $n = 2$, we have $36^1 - 5^2 = 11$. We claim that 11 is the smallest value.

Let $36^m - 5^n = k$. If $2 \mid k$, then $-5^n \equiv 0 \pmod{2}$. This is a contradiction. Similarly, k cannot be divisible by 3 or 5.

If $k = 1$, then we have $1 - 5^n \equiv 1 \pmod{7}$. This yields $-5^n \equiv 0 \pmod{7}$, which is impossible.

If $k = 7$, then we have $1 \equiv 2 \pmod{5}$. This is a contradiction.

Thus, the smallest positive integer of the form $36^m - 5^n$ is 11.

25. Suppose there exist positive integers a, b such that $(15a + b)(a + 15b)$ is a power of 3. Then $15a + b = 3^m$ and $a + 15b = 3^n$ for some positive integers m, n . Taking their sum, we obtain

$$16a + 16b = 3^m + 3^n.$$

The left-hand side is a multiple of 8. As $3^x \equiv 1, 3 \pmod{8}$ for any integer x , the right-hand side can only be congruent to 2, 4, 6 $\pmod{8}$. This is a contradiction. So such a pair of a, b does not exist.

26. Since $x^2 \equiv 0, 1 \pmod{3}$ for any integer x and $3^n \equiv 0 \pmod{3}$, it is not possible to have $p \equiv 2 \pmod{3}$. If $p = 3$, then $3 \mid 3^n + 3$ but $9 \nmid 3^n + 3$. Hence, it suffices to consider the case $p \equiv 1 \pmod{3}$.

Note that

$$\begin{aligned} 3^2 + 7 &= 4^2, \\ 3^1 + 13 &= 4^2, \\ 3^4 + 19 &= 10^2, \\ 3^3 + 37 &= 8^2. \end{aligned}$$

It remains to consider $p = 31$ and $p = 43$. Both of them satisfy $p \equiv 3 \pmod{4}$. If n is odd, then

$$3^n + p \equiv 3 + 3 \equiv 2 \pmod{4}.$$

This cannot be a perfect square since $x^2 \equiv 0, 1 \pmod{4}$ for any integer x . If $n = 2k$ where $k \geq 1$, then

$$p = m^2 - 3^{2k} = (m - 3^k)(m + 3^k)$$

for some m . As p is a prime, this implies $m - 3^k = 1$ and $m + 3^k = p$. This gives $p = 2 \times 3^k + 1$. As $p \leq 50$, we only need to check $k = 1, 2$, and we find that p cannot be 31 or 43.

Hence, the only possibilities are $p = 7, 13, 19, 37$.

27. Since the products of the pairs (a, d) and (b, c) are the same and the difference of the former pair is larger, we have $a + d > b + c$ so that $k > m$. From $ad = bc$, we have

$$a(2^k - a) = b(2^m - b).$$

Considering modulo 2^m , we obtain $-a^2 \equiv -b^2 \pmod{2^m}$. Hence, $2^m \mid a^2 - b^2$, i.e. $2^m \mid (a - b)(a + b)$. Note that $m \geq 1$. Thus, a and b have the same parity. Since

$$(a - b, a + b) = (a - b, 2a) = 2,$$

one of $a - b$ and $a + b$ is even but is not divisible by 4, while the other is divisible by 2^{m-1} . Thus, $a \equiv \pm b \pmod{2^{m-1}}$.

As $2b < b + c = 2^m$, we have $b < 2^{m-1}$ and also $a < 2^{m-1}$. Therefore, the only possibility is $a = 2^{m-1} - b$. This implies

$$c = 2^m - b = 2^m - (2^{m-1} - a) = 2^{m-1} + a.$$

The equation $ad = bc$ becomes

$$a(2^k - a) = (2^{m-1} - a)(2^{m-1} + a).$$

This yields $a \mid 2^{2m-2}$. As a is odd, we must have $a = 1$.

Size

28. Suppose we have $x! + y! = a! + b!$ where $x \leq y$ and $a \leq b$. WLOG we may assume $a \leq x$. If $a = x$, then $b! = y!$. This implies $b = y$ as they are natural numbers.

If $a < x$, then we have $a! + b! = x! + y! \equiv 0 \pmod{x!}$ since $y \geq x$. Note that we cannot have $b \geq x$, since otherwise we have $a! \equiv 0 \pmod{x!}$. This is impossible as $a < x$. Hence, we have $b < x$. However, we find that

$$x! + y! = a! + b! \leq (x-1)! + (x-1)! \leq x!$$

since $x \geq 2$ (from $x > a \geq 1$). This is also impossible.

Therefore, any integer z has at most one such representation.

29. Let k be the integer for which $10^k \leq n < 10^{k+1}$. Then $n^2 < 10^{2k+2}$ so that n^2 has at most $2k + 2$ digits. This yields $S(n^2) \leq 9(2k + 2)$. Thus, we need

$$10^k \leq n = S(n^2) \leq 9(2k + 2).$$

By induction, it is easy to prove that $10^k > 9(2k + 2)$ for $k \geq 2$.

Next, note that $n = S(n^2) \equiv n^2 \pmod{9}$. This implies $n \equiv 0, 1 \pmod{9}$.

For $k = 0$, we have $n = 1, 9$. We check that both satisfy $S(n^2) = n$.

For $k = 1$, we have $n = S(n^2) \leq 9 \times 4 = 36$. It suffices to check $n = 10, 18, 19, 27, 28, 36$. None of these is a solution.

Therefore, the only solutions are $n = 1, 9$.

30. Let k be the integer for which $10^k \leq x < 10^{k+1}$. Then x has $k + 1$ digits so that the product of digits is at most 9^{k+1} . If $k \geq 1$, then this implies

$$9^{k+1} \geq x^2 - 10x - 22 = x(x - 10) - 22 \geq 10^k(10^k - 10) - 22 = 10^{2k} - 10^{k+1} - 22.$$

If $k \geq 2$, then

$$10^{2k} \geq 10 \cdot 10^{k+1} > 10^{k+1} + 10^{k+1} + 10^{k+1} > 10^{k+1} + 9^{k+1} + 22.$$

This is a contradiction. Therefore, we have $k = 0$ or $k = 1$.

If $k = 0$, then x is a unit digit. Thus, $x^2 - 10x - 22 = x$. There is no integer solution as $\Delta = 11^2 - 4(-22) = 209$ is not a square number.

If $k = 1$, then $x^2 - 10x - 22 \leq 9^2$. This implies

$$x \leq \frac{10 + \sqrt{10^2 - 4(-103)}}{2},$$

i.e. $x \leq 16$. Let $x = 10 + a$ for some unit digit a . Then we need $x^2 - 10x - 22 = a$, i.e.

$$(10 + a)^2 - 10(10 + a) - 22 = a.$$

After simplification, this is the same as $(a - 2)(a + 11) = 0$. The only positive solution is $a = 2$, yielding the only answer $x = 12$.

31. Let $n = 1983$. WLOG assume $a \leq b \leq c$. Firstly, we have

$$\frac{1}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{a} + \frac{1}{a} + \frac{1}{a} = \frac{3}{a}.$$

This yields $a \leq 3n$.

Next, the equation trivially implies $\frac{1}{a} < \frac{1}{n}$, i.e. $a \geq n + 1$. Therefore, we have

$$\frac{1}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{n+1} + \frac{1}{b} + \frac{1}{b}.$$

This implies

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \leq \frac{2}{b}.$$

Thus, $b \leq 2n(n+1)$.

For each pair of a and b , there is at most one solution in c . As there are finitely many choices of a and b , there are finitely many solutions.

32. WLOG assume $a \geq b \geq c$. Then

$$2 = \left(1 + \frac{1}{a}\right) \left(1 + \frac{1}{b}\right) \left(1 + \frac{1}{c}\right) \leq \left(1 + \frac{1}{c}\right)^3.$$

If $c \geq 4$, then $\left(1 + \frac{1}{c}\right)^3 \leq \frac{5^3}{4^3} < 2$. This is a contradiction.

If $c = 1$, then $\left(1 + \frac{1}{a}\right) \left(1 + \frac{1}{b}\right) = 1$. This is impossible since a, b are positive.

If $c = 2$, then $\left(1 + \frac{1}{a}\right) \left(1 + \frac{1}{b}\right) = \frac{4}{3}$. After expansion, this becomes $3a + 3b + 3 = ab$, i.e.

$$(a-3)(b-3) = 12.$$

As $a \geq b$, this yields

$$(a-3, b-3) = (12, 1), (6, 2), (4, 3).$$

Correspondingly, we have the solutions $(a, b, c) = (15, 4, 2), (9, 5, 2), (7, 6, 2)$.

If $c = 3$, then $\left(1 + \frac{1}{a}\right) \left(1 + \frac{1}{b}\right) = \frac{3}{2}$. After expansion, this becomes $2a + 2b + 2 = ab$, i.e.

$$(a-2)(b-2) = 6.$$

As $a \geq b$, this yields

$$(a-2, b-2) = (6, 1), (3, 2).$$

Correspondingly, we have the solutions $(a, b, c) = (8, 3, 3), (5, 4, 3)$.

Therefore, the solutions are $(a, b, c) = (15, 4, 2), (9, 5, 2), (7, 6, 2), (8, 3, 3), (5, 4, 3)$ up to permutation.

33. WLOG assume $a \geq b \geq c$. Rewrite the equation as

$$ab + a + b - 1 = (ab - a - b - 1)c.$$

If $c = 1$, then we have $2(a+b) = 0$. There is no solution.

If $c = 2$, then we have $ab - 3a - 3b - 1 = 0$. This yields $(a-3)(b-3) = 10$. As $a \geq b$, we have

$$(a-3, b-3) = (5, 2), (10, 1).$$

This gives the solutions $(a, b, c) = (8, 5, 2), (13, 4, 2)$.

If $c \geq 3$, then since $ab - a - b - 1 = (a - 1)(b - 1) - 2 > 0$, we get

$$ab + a + b - 1 = (ab - a - b - 1)c \geq 3(ab - a - b - 1).$$

This yields $2ab - 4a - 4b - 2 \leq 0$, i.e. $(a - 2)(b - 2) \leq 5$.

If $b = 3$, then we must have $c = 3$. The equation becomes $2a = 14$, i.e. $a = 7$.

If $b \geq 4$, then $a - 2 \leq 2$ and hence $a = b = 4$. The equation becomes $7c = 23$, which has no solution.

Therefore, the solutions are $(a, b, c) = (8, 5, 2), (13, 4, 2), (7, 3, 3)$ up to permutation.

34. No. WLOG assume $x \geq y$. Since $x^2 < x^2 + y \leq x^2 + x < (x + 1)^2$, $x^2 + y$ lies between two consecutive perfect squares. Therefore, it cannot be a square.

35. Note that we have

$$(n + 2)^2 = n^2 + 4n + 4 < n^2 + 12n + 8 < n^2 + 12n + 36 = (n + 6)^2.$$

Therefore, $n^2 + 12n + 8$ can only be $(n + 3)^2$, $(n + 4)^2$ or $(n + 5)^2$. Solving each of these, we find that only $n^2 + 12n + 8 = (n + 4)^2$ yields a solution $n = 2$.

36. The answer is $n = 1, 3$.

We have

$$n^4 - 4n^3 + 22n^2 - 36n + 18 < (n^2 - 2n + 9)^2$$

by expanding the right-hand side. If

$$n^4 - 4n^3 + 22n^2 - 36n + 18 = (n^2 - 2n + 8)^2,$$

then we have $2n^2 - 4n - 46 = 0$. This has no integer solution. In order that the expression is a perfect square, we must have

$$n^4 - 4n^3 + 22n^2 - 36n + 18 \leq (n^2 - 2n + 7)^2.$$

This is the same as $4n^2 - 8n - 31 \leq 0$. Hence,

$$n \leq \frac{8 + \sqrt{8^2 - 4(4)(-31)}}{2(4)} < 4.$$

It suffices to check $n = 1, 2, 3$. When $n = 1$, we find that

$$n^4 - 4n^3 + 22n^2 - 36n + 18 = 1$$

is a perfect square. When $n = 3$,

$$n^4 - 4n^3 + 22n^2 - 36n + 18 = 81$$

is a perfect square. And it is not when $n = 2$. So $n = 1, 3$.

37. WLOG assume $p \leq q$. Let $p^2 + q^3 = m^2$. This implies

$$q^3 = (m - p)(m + p).$$

As q is a prime and $m - p < m + p$, there are two possibilities

$$(m - p, m + p) = (1, q^3), (q, q^2).$$

If $m - p = 1$ and $m + p = q^3$, then $p = \frac{q^3 - 1}{2}$. As $p \leq q$, we have $q^3 - 1 \leq 2q$. However, as $q \geq 2$, we have

$$q^3 - 2q - 1 \geq 4q - 2q - 1 = 2q - 1 \geq 3.$$

This is a contradiction.

If $m - p = q$ and $m + p = q^2$, then $p = \frac{q^2 - q}{2}$. As $p \leq q$, we have $q^2 - q \leq 2q$. This yields $q - 1 \leq 2$, and so the only possibilities are $q = 2, 3$. One checks that only $q = 3$ yields a solution $(p, q) = (3, 3)$, where $p^2 + q^3 = q^2 + p^3 = 36$ is a perfect square.

38. As $4(a^9 + 1)$ and $4(a^3 + 1)$ are perfect cubes, their ratio $\frac{4(a^9 + 1)}{4(a^3 + 1)} = a^6 - a^3 + 1$ is also a perfect cube. Note that

$$a^6 - a^3 + 1 < (a^2)^3$$

if $a > 1$. Also, we claim that

$$a^6 - a^3 + 1 > (a^2 - 1)^3.$$

Indeed, this is equivalent to $3a^4 - a^3 - 3a^2 + 2 > 0$. This is true since

$$3a^4 - a^3 - 3a^2 + 2 \geq 6a^3 - a^3 - 3a^2 + 2 = 5a^3 - 3a^2 + 2 > 2a^2 + 2 > 0.$$

As $a^6 - a^3 + 1$ lies between two consecutive cubes, it cannot be a perfect cube. So the only possibility is $a = 1$. In that case, $4(1^n + 1) = 8$ for any $n \geq 1$. This is a perfect cube. Thus, $a = 1$.

Greatest Common Divisor

39. Since $(m, m + 1) = (m, 1) = 1$, we must have $m = a^n$ and $m + 1 = b^n$ for some positive integers a, b . This shows there are two consecutive positive integers both of which are perfect n th powers. This is impossible.

40. Since $(m, m^2 + 8m + 1) = (m, 1) = 1$, we must have $m = a^2$ and $m^2 + 8m + 1 = b^2$ for some positive integers a, b . The latter equation implies $(m + 4)^2 - 15 = b^2$. This yields

$$(m + 4 - b)(m + 4 + b) = 15.$$

As $m + 4 + b > m + 4 - b$, the only possibilities are

$$(m + 4 - b, m + 4 + b) = (1, 15), (3, 5).$$

This yields $m = 4$ since m is positive. In that case, we have $n^2 = 4 \times 49$ so that $n = 14$. The only solution is $(m, n) = (4, 14)$.

41. Rewrite the equation as $(m^2 - 1)(m^2 + 1) = 2n^2$. Obviously, m is odd. Therefore, we have

$$(m^2 - 1, m^2 + 1) = (m^2 - 1, 2) = 2.$$

Since their product is two times a square number and $4 \mid m^2 - 1$, we must have

$$m^2 - 1 = 4a^2 \quad \text{and} \quad m^2 + 1 = 2b^2$$

for some relative prime nonnegative integers a, b .

From $m^2 - 1 = 4a^2$, we have $(m - 2a)(m + 2a) = 1$. Hence, $m - 2a = m + 2a = 1$. This yields $(m, n) = (1, 0)$, which is a solution by checking.

42. The equation can be rewritten as $m^2 = n(mn - 1)$, i.e.

$$m^3 = (mn)(mn - 1).$$

Since $(mn, mn - 1) = (mn, 1) = 1$, we must have $mn = a^3$ and $mn - 1 = b^3$ for some integers a, b . As mn and $mn - 1$ are two consecutive integers, the only possibilities are $(a, b) = (0, -1), (1, 0)$. In any case, we must have $m = 0$, and hence $n = 0$. Obviously, $(m, n) = (0, 0)$ is a solution.

43. The equation can be rewritten as

$$2^n = m^2 - 49 = (m - 7)(m + 7).$$

Thus, each of $m - 7$ and $m + 7$ is a power of 2. Clearly, m is odd. This yields $(m - 7, m + 7) = (m - 7, 14) = 2$. Therefore, we must have $m - 7 = 2$, i.e. $m = 9$. In that case, $n = 5$.

So the only solution is $(m, n) = (9, 5)$.

44. If $n \leq 7$, then n is even. Indeed, suppose $n = 2m + 1$ is odd. Since $2^{2m}(2 + 2^{8-2m} + 2^{11-2m})$ is a perfect square, so is $2 + 2^{8-2m} + 2^{11-2m}$. However, it is congruent to 2 modulo 4, which cannot be true for a square number. Checking $n = 2, 4, 6$, one finds that $2^8 + 2^{11} + 2^n$ is not a square in these cases.

If $n \geq 8$, then $2^8(2^{n-8} + 9)$ is a perfect square and hence $2^{n-8} + 9$ is a perfect square. Let $2^{n-8} + 9 = k^2$. This is the same as

$$(k - 3)(k + 3) = 2^{n-8}.$$

Thus, each of $k - 3$ and $k + 3$ is a power of 2. Since $(k - 3, k + 3) = (k - 3, 6)$ divides 6, we can only have $k - 3 = 1, 2$. The case $k - 3 = 1$ is rejected. For $k - 3 = 2$, we have $n = 12$. This is a solution since $2^{12-8} + 9 = 5^2$. Thus the only solution is $n = 12$.

45. Suppose on the contrary that $33 \cdots 3 = a^b$ where $b \geq 2$. Let there be k digits 3 on the left-hand side. Then

$$10^k - 1 = 3a^b.$$

It is obvious that a is a multiple of 3. Let $a = 3^\alpha c$ where $3 \nmid c$. Then $10^k - 1 = 3^{\alpha b+1} c^b$. As $b \geq 2$, the right-hand side is a multiple of 27. As $10^3 - 1 = 999$ is divisible by 27 while $10^2 - 1 = 99$ and $10 - 1 = 9$ are not, the order of 10 modulo 27 is 3. Hence, $k = 3\ell$ for some $\ell \in \mathbb{N}$. The equation becomes

$$3^{\alpha b+1} c^b = 10^{3\ell} - 1 = (10^\ell - 1)(10^{2\ell} + 10^\ell + 1).$$

Since

$$(10^\ell - 1, 10^{2\ell} + 10^\ell + 1) = (10^\ell - 1, 1 + 1 + 1) = 3$$

and $9 \mid 10^\ell - 1$, we must have $10^\ell - 1 = 3^{\alpha b} u^b$ and $10^{2\ell} + 10^\ell + 1 = 3v^b$ for some integers u and v . For the equation $10^\ell - 1 = (3^\alpha u)^b$, the only solution is $\ell = 1$ by Catalan's conjecture. In that case, $\frac{10^k - 1}{3} = \frac{10^3 - 1}{3} = 333$ is not a perfect power.

Therefore, the integer $33 \cdots 3$ cannot be a perfect power.